Neuro-Symbolic Causal Inference via Language Model Knowledge Extraction and Bayesian Networks

Phase 1: Advanced Machine Learning Baseline

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Abstract

This report documents Phase 1 of a neuro-symbolic causal inference project using the Corr2Cause benchmark. We compare a rule-based baseline with three feature-engineered machine-learning classifiers—Logistic Regression, Linear SVM, and Random Forest—for determining whether a causal hypothesis is valid given a structured premise. The wider pipeline converts candidate relations into a directed acyclic graph (DAG) and then a Bayesian network for probabilistic queries. On the held-out test set, Linear SVM achieves the strongest macro F1 among the ML classifiers (0.5907), while the rule-based model's higher accuracy (0.8184) is accompanied by very poor positive-class recall. We discuss class imbalance, graph-construction assumptions, prior-based CPDs, and the limits of interpreting the resulting network as validated causal knowledge.

1 Introduction

Correlation indicates statistical association, whereas causation concerns the effect of an intervention. A system that identifies associations alone cannot automatically infer causal direction, rule out confounding, or answer intervention questions. The project explores a neuro-symbolic pipeline combining machine-learning classification with graph constraints and Bayesian-network inference. Phase 1 is the AML baseline; transformer-based deep learning is outside the implemented scope and is reserved for Phase 2.

2 Problem Definition and Objectives

Each Corr2Cause example contains a structured premise describing correlations and/or independence relations and a hypothesis making a causal claim. The binary label is 1 for a valid claim and 0 for an invalid claim. Objectives are: (i) audit the dataset, (ii) engineer interpretable features, (iii) compare a rule-based baseline and ML classifiers, (iv) build an acyclic graph from candidate relations, and (v) demonstrate probabilistic queries with a Bayesian network.

3 Dataset and Data Quality

The project reports 205,734 training rows, 1,076 development rows, and 1,162 test rows. The label distribution is imbalanced: label 1 represents 18.5% of train, 12.9% of development, and 15.5% of test. No missing values or within-split exact duplicate texts were reported. Split verification

found no exact premise–hypothesis overlap between splits. It did report 2 premise-only overlaps between train and development and 3 between development and test; these may reflect different hypotheses applied to a shared premise, but should be inspected and documented.

Table 1: Dataset splits and positive-class prevalence

Split	Rows	Label 0	Label 1
Train	205,734	81.5%	18.5%
Development	1,076	87.1%	12.9%
Test	1,162	84.5%	15.5%

4 Feature Engineering

The repository documents 29 handcrafted features spanning structural counts, density ratios, text statistics, causal cue words, hypothesis-type indicators, and variable overlap. Representative features include the number of correlations, number of independencies, independence ratio, number of variables, and hypothesis type. The appeal of these features is interpretability and low computational cost. Their weakness is that they largely encode surface structure and may not capture the full logical semantics of conditional independence and causal entailment. Any learned preprocessing (for example, scaling or vectorisation) must be fitted on training data only.

5 Models and Evaluation

We compare a conservative regex/pattern-based RuleBased baseline with Logistic Regression, Linear SVM, and Random Forest. The reported metrics include accuracy, macro precision, macro recall, macro F1, and per-class scores. Macro F1 is particularly important because the label distribution is imbalanced.

Table 2: Reported test-set performance

Model	Accuracy	Macro P	Macro R	Macro F1
Logistic Regression	0.6799	0.5934	0.6654	0.5839
Linear SVM	0.6807	0.6014	0.6818	**0.5907**
Random Forest	0.6824	0.5913	0.6601	0.5832
RuleBased	**0.8184**	0.4829	0.4956	0.4725

Linear SVM has the best macro F1 among the ML models. Although RuleBased has the highest accuracy, its class-1 recall is only 0.0278, so it identifies very few valid claims. This illustrates why accuracy alone is inadequate. For Linear SVM, class-1 precision is 0.2815, recall is 0.6833, and F1 is 0.3987. The model retrieves a useful portion of positive examples but produces many false positives.

6 Methodology: Graph Construction

Candidate triples are converted into a directed graph. Self-loops are rejected, and cycles are detected and resolved by removing an edge from each cycle. The repository's cycle-resolution report records six nodes and 30 edges before resolution, with 15 edges removed to obtain a 15-edge DAG. However, the listed confidence values are all 1.000. Therefore, the stated "remove the lowest-confidence edge" strategy cannot distinguish among edges in these cycles;

the implementation's tie-breaking rule (last edge discovered) can affect the resulting graph. The output is acyclic, but acyclicity is a structural property and does not establish that every edge is causally true.

7 Bayesian Network

A Bayesian network pairs a DAG G with conditional probability distributions. Its joint distribution factorises as

$$P(V_1, \ldots, V_n) = \prod_{i=1}^{n} P(V_i \mid \mathrm{Pa}(V_i)).$$

The project uses binary variables and informative conditional probability distributions/priors. Since the abstract variables do not have observational data from which to estimate real-world mechanisms, these CPDs should be interpreted as modelling assumptions for demonstration rather than empirical causal effect estimates. Observational conditioning $P(Y \mid X = x)$ is distinct from intervention $P(Y \mid \mathrm{do}(X = x))$; an intervention represents setting X externally and requires a defensible causal graph and mechanisms.

8 Discussion

The pipeline demonstrates how an interpretable AML classifier can be connected to symbolic graph constraints and probabilistic inference. The best ML result is modest (macro F1 0.5907), and valid claims remain difficult to classify. The class imbalance, abstract variable names, and template-generated language constrain external validity. In addition, per-example hypothesis classification and the creation of a single global causal graph are different tasks; the project should clearly explain how predictions are aggregated into candidate edges and which data support that graph. A DAG and a consistent Bayesian network do not by themselves validate the causal assumptions.

9 Limitations and Future Work

- Evaluate positive-class precision/recall and macro F1 alongside accuracy; consider threshold selection or class weighting on development data.
- Inspect the five premise-only overlaps and report whether they represent intentional shared structures.
- Replace equal edge confidences with calibrated confidence scores and use a documented global graph-aggregation strategy.
- Validate the Bayesian network and intervention queries against known ground-truth structures/effects before making causal claims.
- In Phase 2, fine-tune a transformer such as DistilBERT for premise–hypothesis classification and compare on the same held-out test split.

10 Conclusion

Phase 1 establishes a reproducible AML baseline for structured causal-claim classification and illustrates a route from extracted relations to DAGs and Bayesian networks. Linear SVM is the strongest ML model by macro F1 in the reported results. The next steps are to strengthen positive-class performance, make graph construction more principled, and validate probabilistic queries against ground truth before extending the system with deep learning.

References

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